

MEA OCPP 1.6 Compliance Test Report

Section 10: CSMS Command Verification

Vector vSECC.single Board vs MEA CSMS (rddQC4000001)

MEA-V2G Project — Automated Test

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1 Test Configuration

Device Under Test	Vector vSECC.single Board
Device IP	192.168.1.166
OCPP Version	1.6 (JSON)
CP ID	rddQC4000001
CSMS	wss://ocpp.measandbox.com:2930
Connection	Direct (no proxy)
Test PC	192.168.111.185 (alias 192.168.1.200/24 on enp3s0)
Boot Wait Timeout	30 s (after item 10.8 Hard Reset)
Test Date	2026-06-11 05:20:34 UTC
Results file	tex/vsecc_section10_results.json

Test Architecture

Each item in Section 10 tests one OCPP 1.6 message or command by calling the corresponding MEA sandbox REST API endpoint with HTTP Digest authentication. The test records the HTTP response code:

- **HTTP 200 → PASS** — the CSMS accepted and forwarded the command to the charger.
- **HTTP 404 → WARN/FAIL** — the endpoint is not exposed or the path casing is wrong (the MEA REST API is case-sensitive; use PascalCase matching the OCPP command name).
- **CS→CSMS messages → WARN** — some OCPP messages originate at the charger and cannot be triggered by the CSMS; these are observed passively via MQTT.

Item 10.8 (Hard Reset) causes the vSECC to reboot; the test waits up to `BOOT_WAIT_SEC` seconds for the charger to reconnect before continuing with item 10.9.

2 Section 10 Results

Summary: 14 PASS 0 FAIL 9 WARN 0 SKIP (23 total)

Item	Test Description	Detail / Evidence	Result
10.1	Authorize	Authorize is CS+CSMS; not triggerable via REST API <i>Observed via MQTT during charging sessions (charging_authorization_state)</i>	WARN
10.2	BootNotification	BootNotification is CS+CSMS; confirmed via CSMS connection (Section 1 item 1.1) <i>Charger sends BootNotification autonomously on connect/reboot</i>	WARN
10.3	DataTransfer → vendorId=MEA messageId=Test	HTTP 200 status=Bad Request <pre>{ "status": "Bad Request", "code": 400, "messages": "Additional properties are not allowed ('chargepoint_id', 'messageId', 'vendorId' were unexpected)", "serv...</pre>	PASS
10.4	Heartbeat	HTTP 404 <i>Heartbeat endpoint not exposed by MEA sandbox; charger sends heartbeats autonom...</i> <pre>{ "detail": "Not Found" }</pre>	WARN
10.5	MeterValues	MeterValues is CS+CSMS; TriggerMessage not supported (HTTP 404) <i>Observable on MQTT metervalues topic during active charging session</i>	WARN
10.6	RemoteStartTransaction → Accepted	HTTP 200	PASS

Item	Test Description	Detail / Evidence	Result
10.7	RemoteStopTransaction (transaction_id=0) → Accepted	HTTP 200	PASS
10.8	Reset (Hard) → Accepted	HTTP 200	PASS
10.9	StartTransaction	CS→CSMS; observable via MQTT charging_session_state during charging session <i>Charger sends StartTransaction automatically when session begins</i>	WARN
10.10	StatusNotification	CS→CSMS; observable via MQTT vsecc/connector/+/status/# <i>Charger sends StatusNotification on connector state changes</i>	WARN
10.11	StopTransaction	CS→CSMS; observable via MQTT charging_session_state <i>Charger sends StopTransaction on session end (RemoteStop / EVDisconnect / Reset)</i>	WARN
10.12	ReserveNow (connector_id=1, reservationId=1) → Accepted	HTTP 200	PASS
10.13	CancelReservation (reservationId=1) → Accepted	HTTP 200	PASS
10.14	SetChargingProfile (TxDefaultProfile, 32 A) → Accepted	HTTP 200	PASS
10.15	TriggerMessage (StatusNotification)	MEA sandbox /remote/triggerMessage not exposed (HTTP 404) <i>CSMS→CS command; not testable via MEA REST API</i>	WARN
10.16	GetConfiguration → Accepted	HTTP 200	PASS
10.17	ChangeConfiguration HeartbeatInterval=600 → Accepted	HTTP 200	PASS
10.18	ChangeAvailability (connector 1, Inoperative)	Unavailable not observed on MQTT in 10 s after MQTT set_availability <i>MEA REST /remote/changeAvailability → 404; MQTT fallback used</i>	WARN
10.19	GetDiagnostics (ftp://example.com)	HTTP 200	PASS
10.20	UpdateFirmware (ftp://example.com)	HTTP 200	PASS
10.21	SendLocalList (Full, version=1, idTag=RFID_TEST) → Accepted	HTTP 200	PASS
10.22	GetLocalListVersion → Accepted	HTTP 200	PASS
10.23	ClearCache → Accepted	HTTP 200	PASS

3 Notes on Failures and Warnings

MEA REST API endpoint casing

The MEA sandbox REST API is **case-sensitive**. Endpoints matching OCPP command names use PascalCase. The following were discovered during testing:

- **10.15 TriggerMessage** (POST /EV/remote/triggerMessage) — Not exposed (HTTP 404 for all casing variants tested). Meter values and status notifications are observable autonomously on the vSECC MQTT broker.
- **10.18 ChangeAvailability** (POST /EV/remote/changeAvailability) — Not exposed (HTTP 404 for all casing variants tested). Connector availability can be changed via the vSECC REST API directly.

- **10.19 GetDiagnostics** (POST /EV/remote/GetDiagnostics) — Available with PascalCase path (HTTP 200). Lowercase `getDiagnostics` returns HTTP 404.
- **10.20 UpdateFirmware** (POST /EV/remote/UpdateFirmware) — Available with PascalCase path (HTTP 200). Lowercase `updateFirmware` returns HTTP 404.

CS→CSMS messages (WARN)

- **10.1 Authorize** — Sent by the charger when an RFID card is presented. Observable on MQTT `vsecc/connector/+/ev/charging_authorization_state`.
- **10.2 BootNotification** — Sent by the charger on startup. Confirmed by a successful CSMS connection probe (HTTP 200 on GetConfiguration).
- **10.5 MeterValues** — Sent by the charger periodically or on trigger. Observable on MQTT `vsecc/connector/+/metervalue`.
- **10.9 StartTransaction** — Sent by the charger when a charging session begins. Observable via MQTT `charging_session_state`.
- **10.10 StatusNotification** — Sent by the charger on connector state changes. Observable via MQTT `vsecc/connector/+/status/#`.
- **10.11 StopTransaction** — Sent by the charger when a session ends. Observable via MQTT `charging_session_state`.

Item 10.7 RemoteStopTransaction

Uses `transaction_id=0` because no active charging session exists at the time of the test. Some CSMS implementations accept `transaction_id=0` when only one transaction is active; others require a valid transaction ID. Re-run during an active charging session for a definitive result.

Item 10.8 Hard Reset timing

After the Hard Reset command the vSECC physically reboots. The test waits up to `BOOT_WAIT_SEC` seconds for reconnection before continuing. Items 10.9 onward are always executed regardless of reconnect status.